

Kokou Hugues Minasseh

(+44) 7818858205 - mink_hugues@outlook.com - Hull, UK

github.com/Hugues-hash | linkedin.com/in/kokou-hugues-minasseh-b426591b8

Available now | Open to remote, hybrid or on-site across the UK | No visa sponsorship required

SUMMARY

Commonwealth Scholar and MSc Artificial Intelligence & Data Science graduate with an engineering background, who builds and validates machine learning and quantitative models in Python and C++. Work spans deep learning, NLP, time-series forecasting, financial modelling and model validation, alongside statistical analysis and data reporting. Focused on turning complex problems into reliable, well-tested, explainable solutions.

EDUCATION

MSc, Artificial Intelligence & Data Science

Sep 2025 - Sep 2026

University of Hull, UK | Commonwealth Scholar (UK Government / FCDO)

- Taught master's combining coursework across seven modules with a 60-credit research project.
- Research project (dissertation): adaptive machine learning for volatility forecasting in financial time series, using regime-aware training with a two-state Hidden Markov Model, Random Forest and XGBoost, and SHAP for explanation stability.
- Fundamentals of Data Science: cleaning and validating messy real-world data, then drawing conclusions with statistics, hypothesis testing and clear visualisation.
- Big Data and Data Mining: SQL databases, association rule mining, KMeans and DBSCAN clustering, and SARIMA time-series forecasting on a real UK road-accident dataset.
- Applied and Understanding Artificial Intelligence: building and evaluating machine learning and deep learning models across NLP, computer vision, regression and clustering, including transformer models such as FinBERT and CNN image classifiers.
- Programming and Research Methods: Python for data work, plus research design, literature review and ethics.

MSc, Mechatronics Engineering

Sep 2019 - Jul 2022

University of Shanghai for Science and Technology, China

- Focused on computer vision and mobile robotics. Published research on an autonomous human-following robot using RGB-D and LIDAR sensing.

Bachelor, Electrical Engineering

Sep 2015 - Jul 2018

National Higher School of Engineers (ENSI), Lome, Togo

PROJECTS

American Sign Language Recognition (Hackathon Winner) | University of Hull (DAIM)

- Adapted and evaluated a pretrained deep-learning video model (I3D) for American Sign Language recognition using the WLASL100 dataset.
- Performed model evaluation and error analysis using top-k metrics and confusion matrices to identify systematic classification weaknesses.
- Built a prototype for context-aware prediction refinement

Volatility Forecasting & Strategy Backtesting | github.com/Hugues-hash/volatility-strategy-backtesting

- Compared six volatility models (rolling, EWMA, GARCH, GJR, EGARCH, Random Forest) out of sample on 21 years of S&P 500 data, ranked by QLIKE and tested for significance with Diebold-Mariano.
- Built a volatility-targeting strategy and backtested it against buy-and-hold, roughly halving the maximum drawdown while tracking Sharpe, turnover, and trading costs.
- Validated each forecast as a one-day Value-at-Risk model with the Kupiec and Christoffersen tests, under both normal and fat-tailed Student-t assumptions.
- Showed the most accurate forecast, the best strategy, and the best risk model are three different things, which is the core lesson of model validation.

Regime & Drift Detection in Financial Time Series | github.com/Hugues-hash/drift_detection

- Built a regime detection and model monitoring framework on S&P 500 returns using Hidden Markov Models.
- Quantified distribution shifts using PSI, KS test, and KL divergence, showing a clear link between data drift and model prediction error.
- Designed the pipeline to flag when models degrade in production-like conditions, supporting more reliable decision-making.

Monte Carlo Option Pricer (C++) | github.com/Hugues-hash/monte-carlo-option-pricer

- Built an options pricing engine in C++ and validated it against the closed-form Black-Scholes price, confirming Monte Carlo convergence at the theoretical rate.
- Added antithetic and control-variate variance reduction and finite-difference Greeks using common random numbers, matching the analytic Greeks to several decimals.
- Exposed the C++ core to Python through a pybind11 binding and benchmarked it against NumPy and pure Python.

NLP Classification with Deep Learning | University of Hull

- Developed and evaluated text-classification pipelines using Scikit-Learn (Naïve Bayes, SVM) and TensorFlow (LSTM, Transformer), comparing performance across multiple metrics.
- Optimised model accuracy through feature engineering and hyperparameter tuning.

Traffic Accident Data Mining & Time Series Forecasting | University of Hull

- Analysed UK road accident data using spatiotemporal analysis, association rule mining, and clustering to identify patterns across regions.
- Built time-series models to forecast weekly accident counts, combining statistical methods with domain context for actionable insight.

Predictive Modelling & Customer Behaviour Analysis | University of Hull

- Built and compared regression and classification models on structured datasets, evaluating with AUC-ROC and F1-score.

Census Data Analysis & Statistical Decision-Making | University of Hull

- Applied hypothesis testing, imputation, and statistical inference on population data to support infrastructure investment recommendations.

WORK EXPERIENCE

Content Manager

Apr 2026 - Present

Redeployable • Hull, UK (Remote)

- Query and analyse candidate performance data with SQL and build reporting dashboards to surface trends, evaluate assessment effectiveness, and drive continuous improvement of job simulations.

- Design and maintain realistic, role-relevant job simulations, and manage job listings to support candidate-to-role matching across multiple opportunities.

Flight Operation Engineer

Jul 2023 - Aug 2025

ASKY Airlines • Lomé, Togo

- Optimised flight route planning by balancing safety constraints, fuel efficiency, and operational performance using analytical tools.
- Evaluated aircraft performance limitations and fuel requirements, translating complex operational data into practical decisions.

Electrical Engineer

Nov 2022 - Jul 2023

Dorilus Africa • Lomé, Togo

- Designed electrical systems across low and high-voltage projects, ensuring compliance with international standards.
- Performed load analyses and supported solar energy optimisation projects, identifying issues early through systematic data review.

SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn, TensorFlow, arch, hmmlearn, Matplotlib, Seaborn), C++, SQL

Machine Learning: TensorFlow, Scikit-Learn, Supervised & Unsupervised learning, feature engineering, deep learning (LSTM, Transformer), NLP, model evaluation, hyperparameter tuning, time-series forecasting.

Statistical Analysis: Hypothesis testing, distribution analysis, feature engineering, predictive modelling, drift & regime detection

Quantitative Finance: GARCH/GJR/EGARCH, Monte Carlo simulation, Value at Risk (Kupiec, Christoffersen), Hidden Markov regime detection, backtesting (Sharpe, drawdown, turnover).

Tools: Jupyter Notebook, Git, CMake, pybind11

PUBLICATION

Minasseh, K.H. (2021). *An autonomous human-following Robot using RGB-D camera and 2D LIDAR*. High Technology Letters, 27(12).

LANGUAGES

English (Fluent) | French (Native)